

Frontend – Assignment 1

Answers

Q1. HTTP Request–Response Cycle

The HTTP Request–Response Cycle is the process through which a browser communicates with a web server.

When a user enters a URL, the browser sends an HTTP request to the server. The server processes the request and sends an HTTP response back. The browser receives the response and displays the webpage.

Main steps:

1. User enters a URL or clicks a link.
2. Browser sends an HTTP request to the web server.
3. Server processes the request.
4. Server sends an HTTP response.
5. Browser receives the response and displays the webpage.

Simple diagram:

Browser → HTTP Request → Web Server

Browser ← HTTP Response ← Web Server

Q2. Introduction to Web Technologies

1. HTML

HTML is used to create the basic structure of a webpage. It is used for headings, paragraphs, images, links, tables, forms, and lists.

2. CSS

CSS is used to style a webpage. It controls colors, fonts, spacing, borders, layout, and responsive design.

3. JavaScript

JavaScript adds functionality and interaction to a webpage. For example, it can handle button clicks, validate forms, and change webpage content.

4. Bootstrap

Bootstrap is a frontend framework with ready-made classes and components. It helps developers create responsive websites faster.

5. jQuery

jQuery is a JavaScript library that makes tasks like DOM manipulation, events, animations, and AJAX easier.

6. React.js

React.js is a JavaScript library used to create user interfaces. It uses reusable components, making interactive webpages and applications easier to build.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies work mainly in the user's browser. They create the visible part of a website and handle interactions on the user's device.

Examples: HTML, CSS, JavaScript, and React.js.

Server-side technologies work on the web server. They handle requests, application logic, authentication, and database operations.

Examples: Node.js, PHP, Python, and Java.

Main difference:

- Client-side: Runs in the browser and mainly handles the user interface.
- Server-side: Runs on the server and mainly handles data processing and application logic.

Both work together because the client sends a request to the server and receives a response.

Q4. Block-Level Elements in HTML

Block-level elements are HTML elements that normally start from a new line and take the available width of their parent element.

Characteristics:

- They usually start on a new line.
- They generally occupy the available width.
- They are useful for creating the main structure of a webpage.

Examples:

1. <div> – general container.
2. <p> – paragraph.
3. <h1> – main heading.

Other examples include <section>, <article>, <header>, and <footer>.

Q5. Semantic HTML Elements

Semantic HTML elements have names that clearly describe the purpose of their content. They make a webpage structure easier to understand.

Importance:

- Makes code easier to read and maintain.
- Helps search engines understand page structure.
- Improves accessibility for screen-reader users.

Examples:

<header> – header of a page or section.
<nav> – navigation links.
<main> – main content.
<section> – a particular section.
<article> – independent content.

<footer> – footer information.

Q6. Creating a Table Using HTML

HTML code is given below:

Sr. No.	Name	Designation	Department
1	John	Manager	Sales
2	Smith	Sr. Manager	Marketing
3	Jane	Manager	HR

```
<table border="1">
  <tr>
    <th>Sr. No.</th><th>Name</th><th>Designation</th><th>Department</th>
  </tr>
  <tr><td>1</td><td>John</td><td>Manager</td><td>Sales</td></tr>
  <tr><td>2</td><td>Smith</td><td>Sr. Manager</td><td>Marketing</td></tr>
  <tr><td>3</td><td>Jane</td><td>Manager</td><td>HR</td></tr>
</table>
```

Q7. Creating a Nested List Using HTML

HTML code is given below:

```
<ul>
  <li>River
    <ul>
      <li>Ganga</li><li>Yamuna</li>
      <li>Brahmaputra</li><li>Narmada</li>
    </ul>
  </li>
</ul>
```

Q8. Types of CSS

There are three main types of CSS:

1. Inline CSS

Inline CSS is written directly inside an HTML element using the style attribute.

Example:

```
<p style="color: blue;">Hello</p>
```

2. Internal CSS

Internal CSS is written inside the <style> tag in the <head> section.

Example:

```
<style>
p {
  color: blue;
}
</style>
```

3. External CSS

External CSS is written in a separate .css file and connected using the <link> tag.

Example:

```
<link rel="stylesheet" href="style.css">
```

External CSS is useful for larger websites because the same CSS file can be used for multiple pages.

Q9. Practical Understanding

HTML, CSS, and JavaScript work together to create a complete webpage.

HTML gives the webpage its structure, such as headings, paragraphs, images, buttons, and forms.

CSS controls the appearance of these elements. It is used for colors, fonts, spacing, layout, and responsive design.

JavaScript adds behavior and interaction. It can respond to button clicks, validate forms, perform calculations, and change webpage content.

In simple words:

HTML = Structure

CSS = Design

JavaScript = Functionality and Interactivity

Q10. Short Answer

a. Role of Bootstrap

Bootstrap is a frontend framework that provides ready-made styles and components. It helps create responsive and mobile-friendly websites quickly.

b. What is jQuery?

jQuery is a JavaScript library that simplifies tasks such as DOM manipulation, event handling, animations, and AJAX operations.

c. What is React.js?

React.js is a JavaScript library for building user interfaces. It uses reusable components, which makes interactive frontend applications easier to develop and maintain.